Phishing Email Detection & Awareness Report

Prepared By: Abhishek Singh | Date: 27 February 2026
Project Type: Security Awareness & Email Threat Analysis
Tools Used: Google Header Analyzer, MXToolbox, WHOIS, VirusTotal, Manual Analysis

Executive Summary

This report presents a structured analysis of phishing emails identified during security evaluation. The objective is to detect malicious characteristics, assess technical authentication failures, and provide employee awareness guidance to reduce organizational risk.

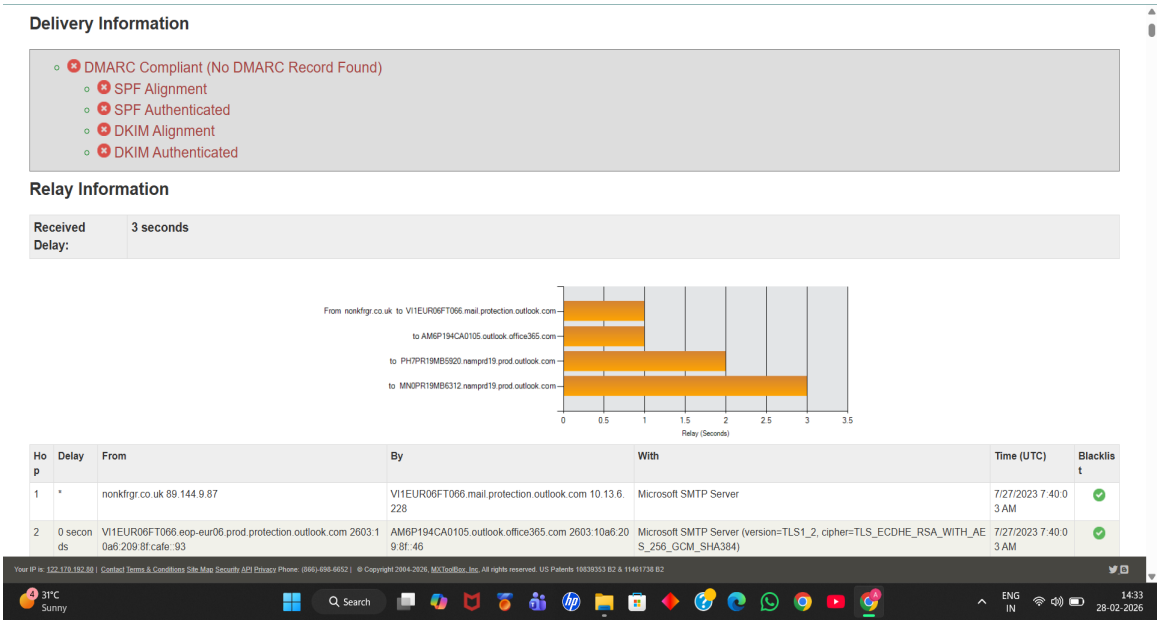
Methodology

Collected suspected phishing emails. Analyzed headers using Google Header Analyzer and MXToolbox. Verified domain reputation using WHOIS and VirusTotal. Inspected URLs manually. Identified social engineering indicators. Classified risk level based on authentication failures and behavioral tactics.

Header Analysis

SPF: Failed or not aligned. DKIM: Not authenticated. DMARC: No record found. Sending domain not authorized to send emails. Originating IP unrelated to claimed organization. Multiple authentication failures indicate high phishing probability.

Header Analysis Evidence (Screenshots)



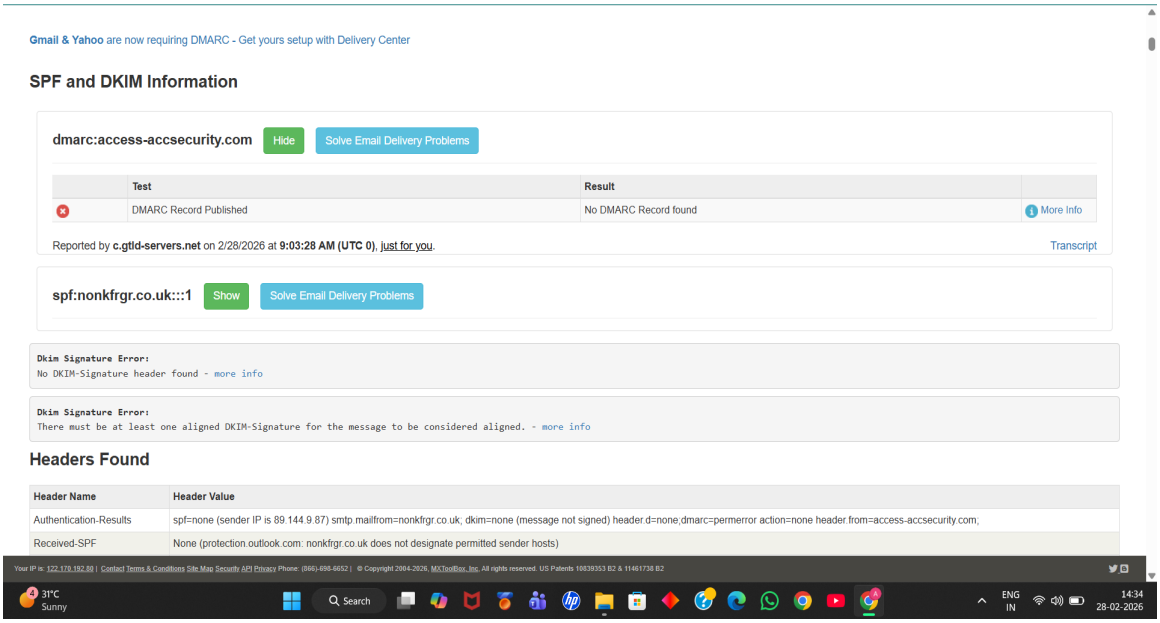
Delivery Information

- DMARC Compliant (No DMARC Record Found)
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

Relay Information

Received Delay:	3 seconds

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	nonkfrgr.co.uk 89.144.9.87	VI1EUR06FT066.mail.protection.outlook.com 10.13.6.228	Microsoft SMTP Server	7/27/2023 7:40:03 AM	✔
2	0 seconds	VI1EUR06FT066.eop-eur06.prod.protection.outlook.com 2603:10a6:209:8f:cafe::93	AM6P194CA0105.outlook.office365.com 2603:10a6:209:8f::46	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	7/27/2023 7:40:03 AM	✔



Evidences

No DMARC record found. SPF alignment failed. DKIM not authenticated. Sender domain mismatch. Suspicious relay path. Recently registered or unverified domain.

Email Analysis Summary

Email 1:
Subject: Urgent: Your Account Will Be Locked
Dear User, We noticed suspicious activity. Verify Now: http://secure-account-verify[.]com
Risk: Phishing (High Risk)

Email 2:
Subject: Invoice Payment Pending
Dear Customer, Please download the attached invoice to avoid penalty.
Attachment: invoice_update.xlsm
Risk: Phishing (Malware Attachment)

Email 3:
Subject: CEO Request – Immediate Transfer
Hi Finance Team, Transfer funds urgently for confidential project.
Risk: Spear Phishing / Business Email Compromise

Phishing Indicator Summary

Generic greeting. Urgency or fear tactics. Suspicious links. HTTP instead of HTTPS. Domain mismatch. Unexpected attachments. Request for credentials or OTP. Authentication failures in header.

Employee Awareness

Employees must verify sender identity, inspect links before clicking, report suspicious emails immediately, and avoid sharing credentials. Awareness reduces successful phishing attempts significantly.

Do's and Don'ts

DO: Verify sender. Hover over links. Enable MFA. Report suspicious emails. Keep systems updated.
DON'T: Click unknown links. Download unexpected attachments. Share passwords or OTPs. Respond to urgency-based emails without verification.

Conclusion

Phishing attacks exploit human behavior rather than system vulnerabilities. Strong authentication controls combined with continuous employee awareness significantly reduce email-based threats.